Hotspot-Induced GPU Imbalance in Irregular Probabilistic Workloads

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Problem + key idea

Why does GPU efficiency degrade?

Probabilistic cellular automata generate spatially evolving hotspots

As tumour cells activate and proliferate, computation becomes increasingly concentrated in specific lattice regions

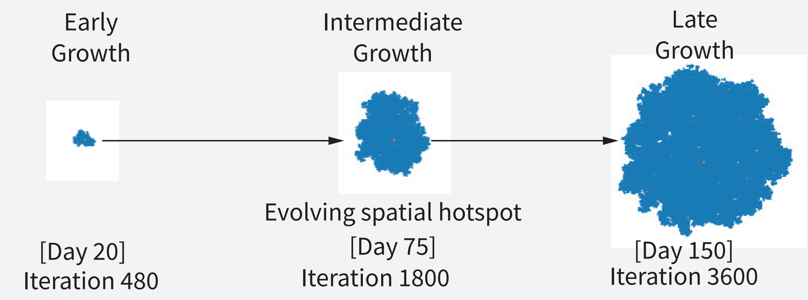
Static GPU mappings then assign very different workloads to thread blocks and SMs, reducing utilisation and execution efficiency.

Tumour growth → Hotspot concentration → Uneven block workloads → SM imbalance

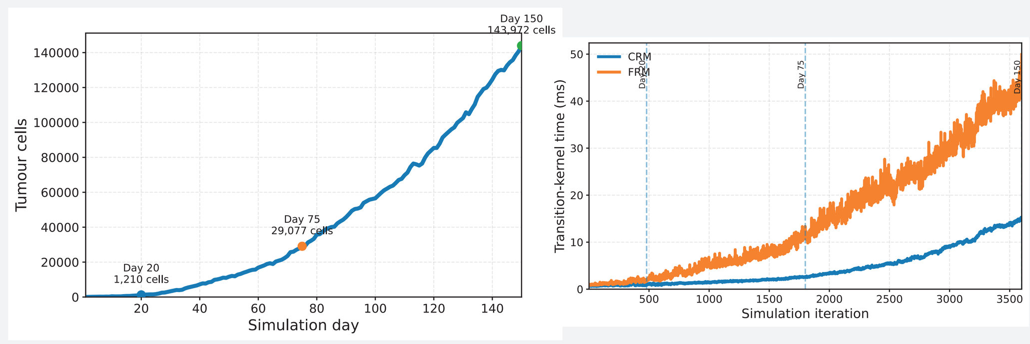
Can we disperse hotspots across the GPU without destroying memory locality?

Same GPU. Same algorithm. Same probabilistic semantics. Different spatial mapping

From tumour growth to GPU execution cost



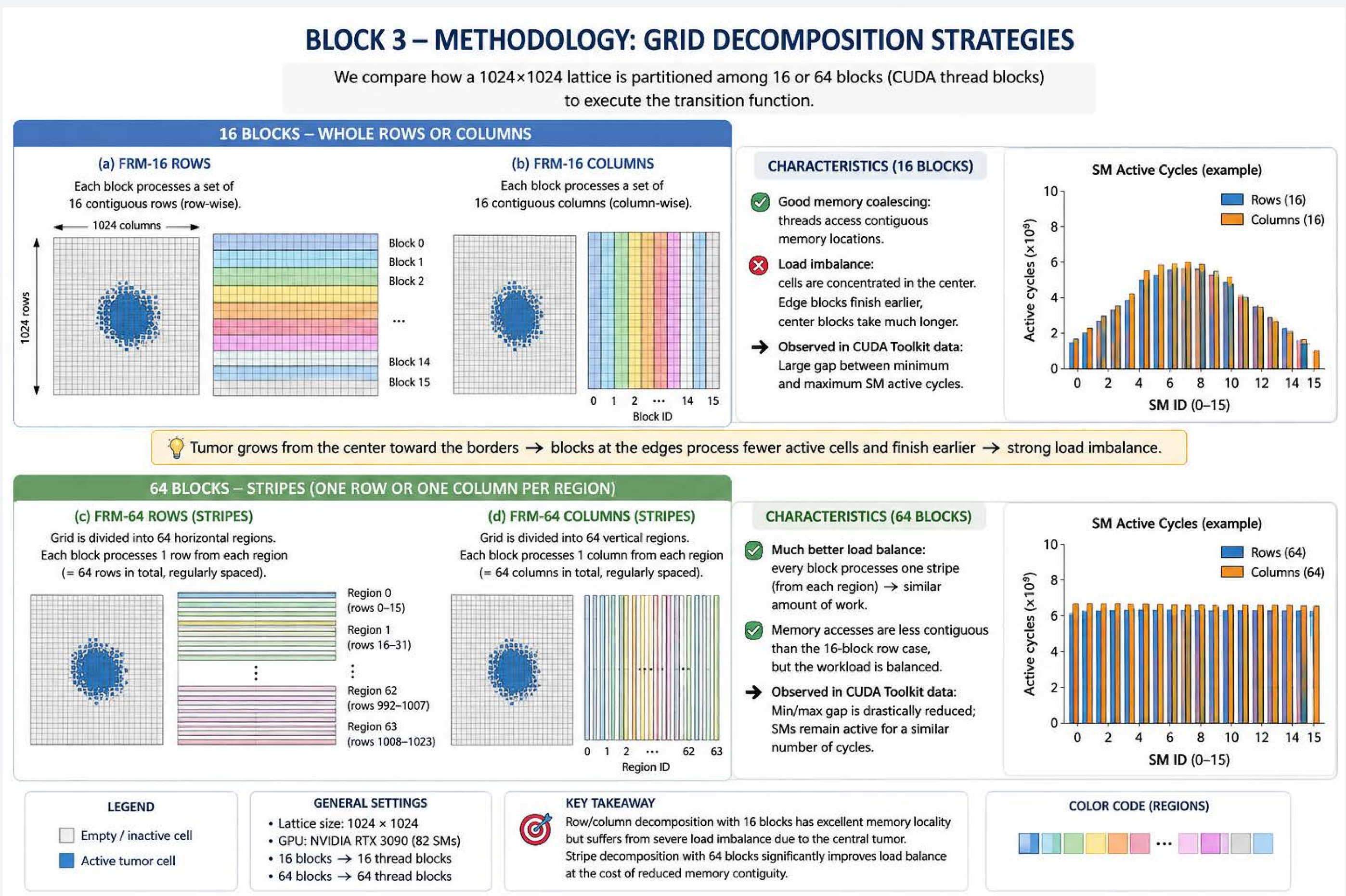
Measured on NVIDIA RTX 3090



- Tumour growth strongly predicts transition-kernel cost $r_{Pearson}>0.99$
- Transition kernel ≈ 81%of total GPU execution time

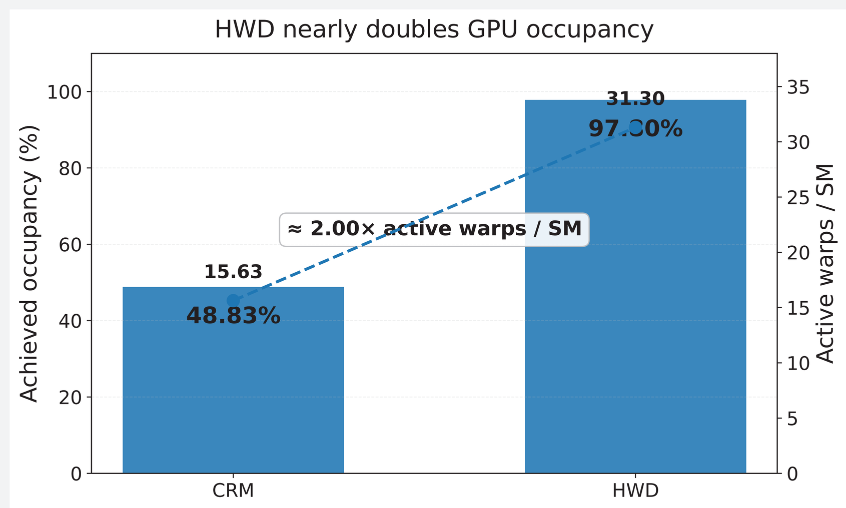
Methodology

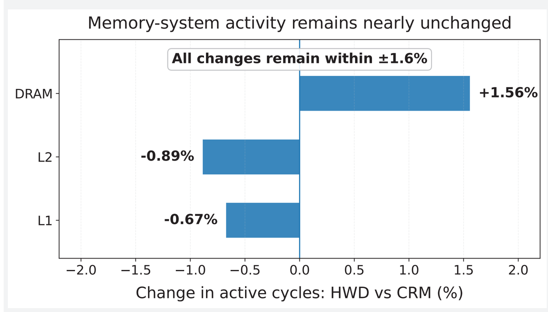
Grid division strategies



We compare how a 1024 ×1024 lattice is partitioned among 16 or 64 blocks (CUDA thread blocks) to execute the transition function kernel

Profiling evidence

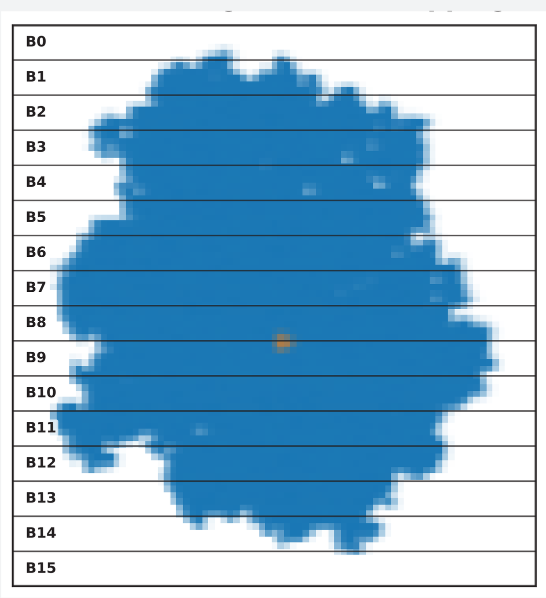




Subsystem	CRM	HWD	HWD vs CRM
L1	435.89M	432.96M	-0.67%
L2	410.79M	407.15M	-0.89%
DRAM	621.93M	631.61M	+1.56%

The gain comes from how computation is distributed, not from reduced memory-system activity.

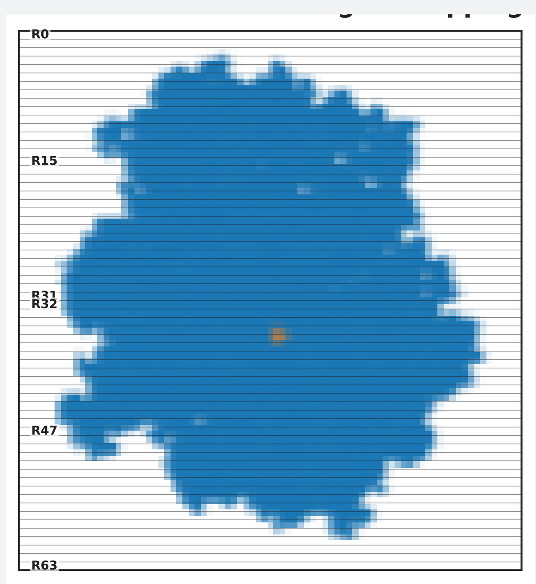
Contiguous Row Mapping (CRM)



16 contiguous horizontal regions

Contiguous regions →Unequal workload
Locality but imbalance

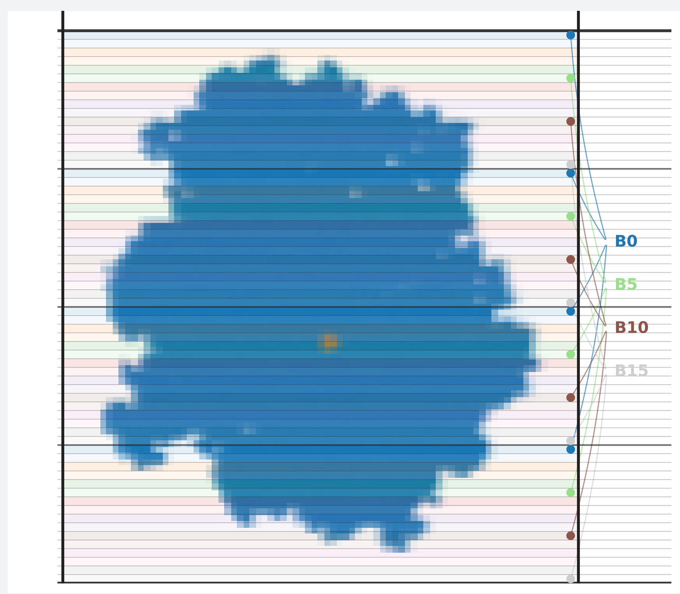
Fine Grained Region Mapping (FRM)



64 horizontal regions

Finer partition →Better distribution

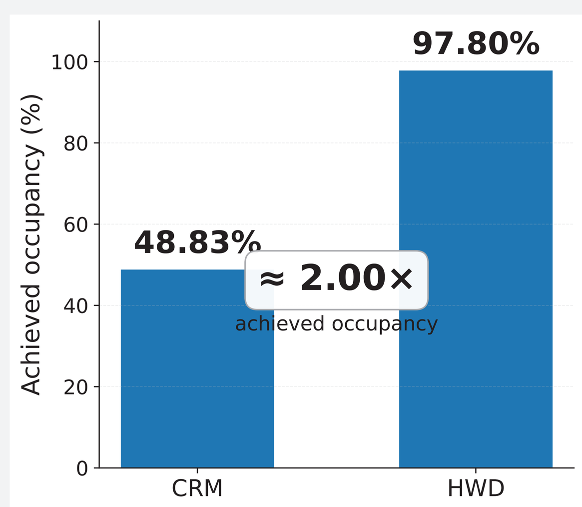
Hybrid Workload Dispersion (HWD)



Spatial dispersion + contiguous access within each region

Distributed hotspot work → more balanced SM use

GPU Occupancy

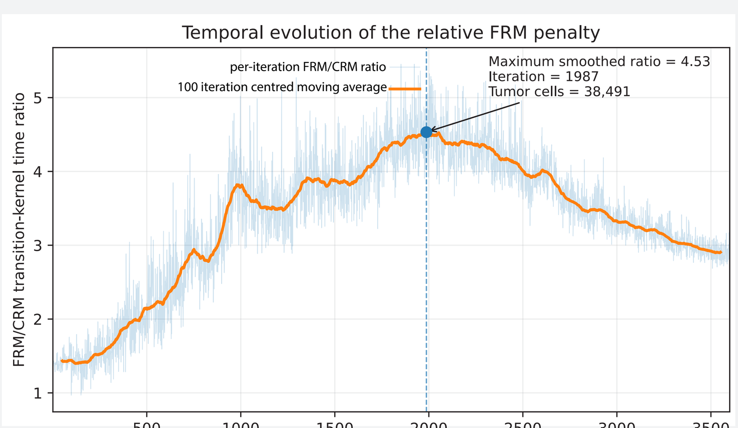


HWD nearly doubles achieved occupancy by balancing work across SMs

Better workload dispersion → higher GPU occupancy

GPU imbalance evolves non-monotonically

FRM / CRM transition kernel-time



- Tumour growth predicts computational cost ($r_{Pearson}>0.99$)
- Tumour cells →Transition-kernel execution time
- Transition ≈ 81% of total execution time

Unexpectedly, the relative FRM overhead decreases again during late tumour growth
Absolute kernel time continues to increase

Conclusions and take-home message

1. **Hotspots matter**
   Probabilistic spatial activation progressively creates heterogeneous GPU workloads
2. **Mapping matters**
   Hybrid workload dispersion raises achieved occupancy from 48.83% to 97.80% while memory-subsystem activity remains essentially unchanged
3. **Mapping should eventually become adaptive**
   The non-monotonic FRM/CRM behaviour shows that no single static mapping is necessarily optimal throughout the simulation

Adaptive mapping driven by runtime hotspot and GPU-imbalance indicators is the next step

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